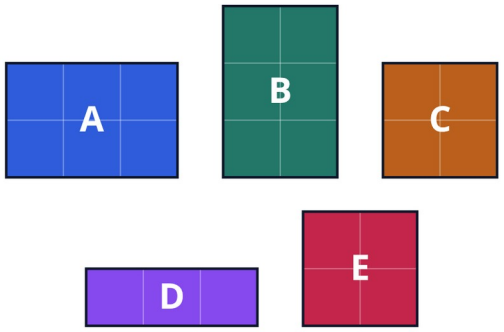
B*-tree floorplan representation

Five blocks, one frame, and a tree that says where each block goes

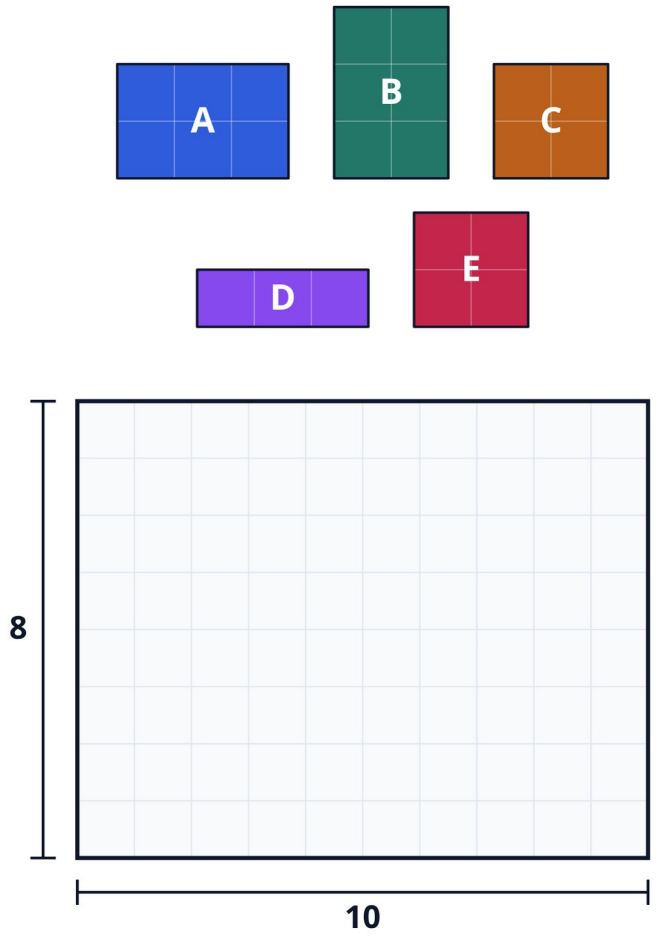
Five blocks

- A chip is made of large rectangular parts, called blocks
- Ours has five blocks, named A to E
- They are drawn on squared paper, so you can count each width and height



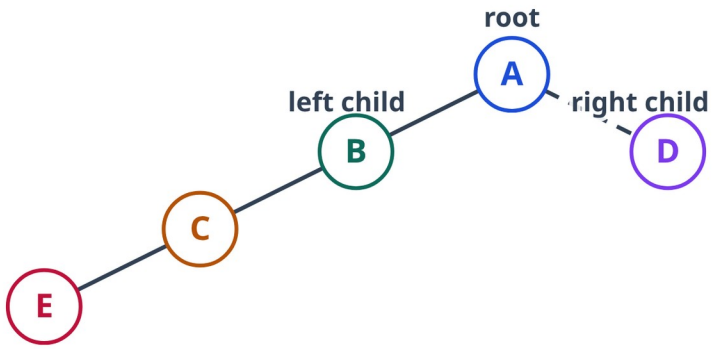
The outline

- A floorplan says where each block sits on the chip
- Every block must stay inside a frame called the outline
- Our outline is ten squares wide and eight squares tall
- The tighter the blocks are packed, the smaller the chip can be



A tree as a recipe

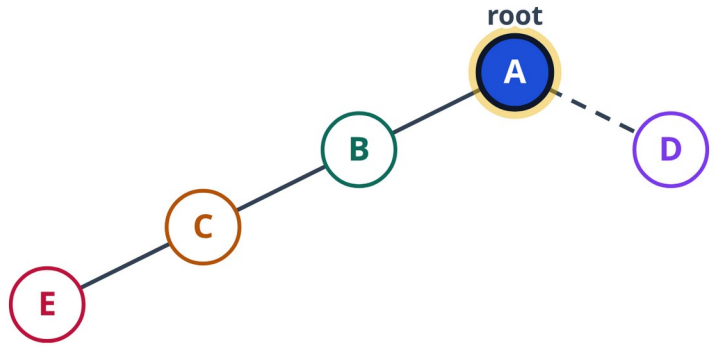
- Writing down every position is slow, so we use a recipe instead: a B-star tree
- Each block is one node of the tree
- The top node, A, is called the root
- Each node can have two children: a left child and a right child

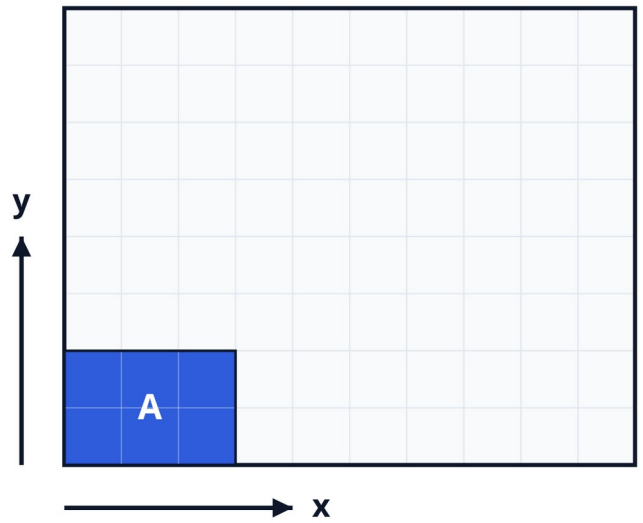




The root goes in the corner

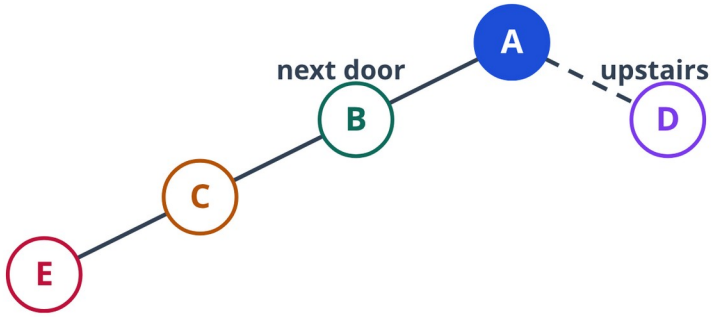
- The root goes in the lower-left corner of the outline
- From that corner, x counts squares to the right, and y counts squares up

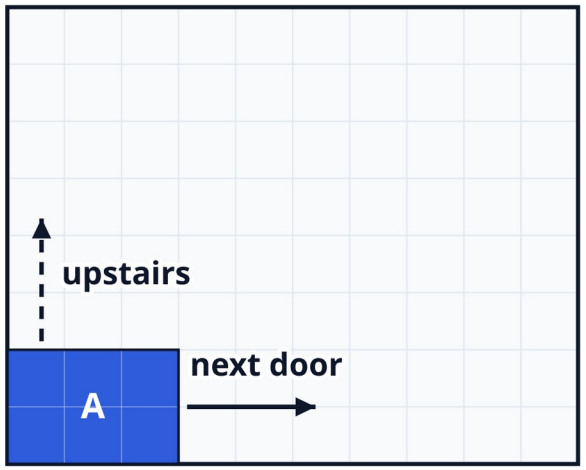




Left is next door, right is upstairs

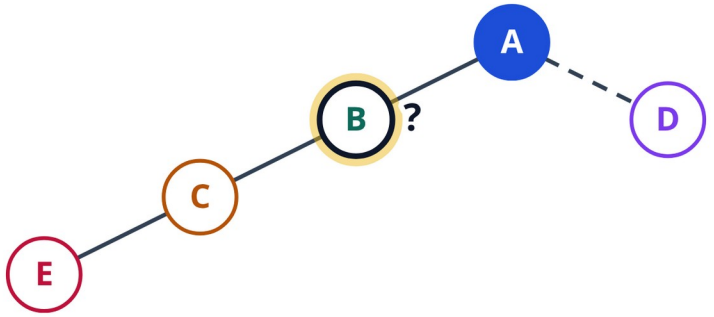
- A left child goes next door, right against its parent's right side
- A right child goes upstairs, on top of its parent, with the same left edge
- The names sound backwards, so remember: left means next door, right means upstairs
- In the tree, a solid line means next door, and a dashed line means upstairs

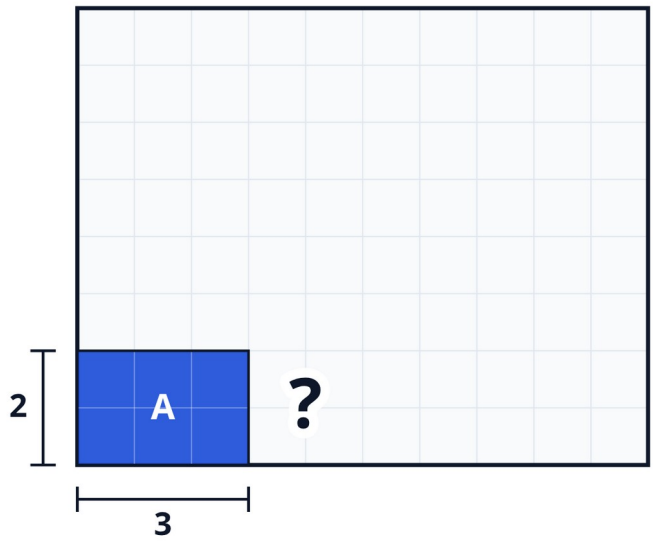




Where does B go?

- A is three squares wide and two squares tall
- B is A's left child
- Pause and guess: where does B's left edge go?



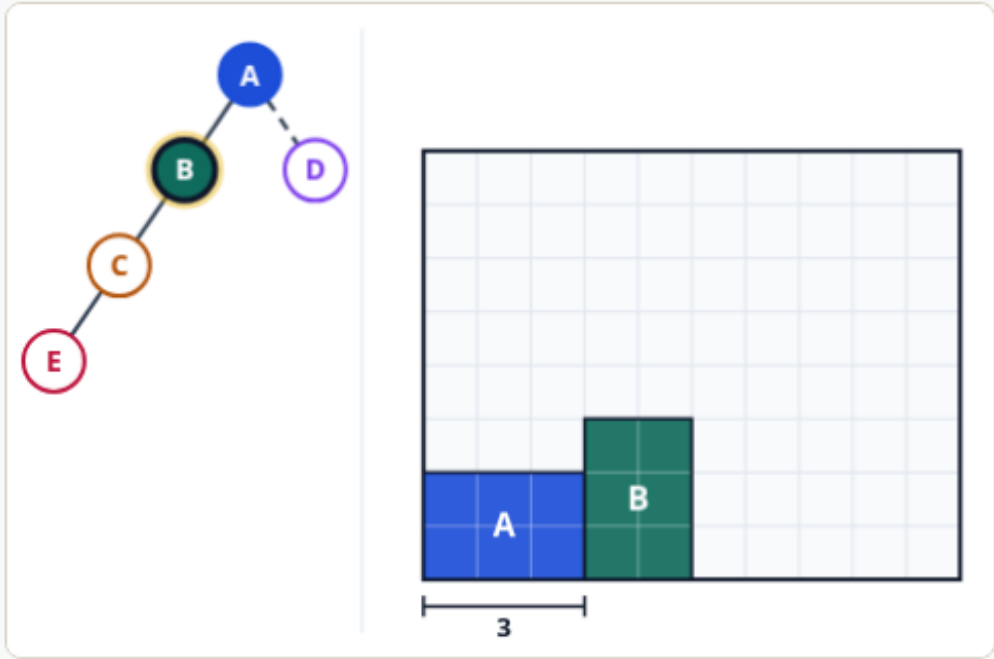


B goes next door, at x = 3

B*-TREE

B goes next door, at x = 3

Step 1 / 5 · root · module02-03-bstar-tree



Explanation

B goes next door, at x equals three: A starts at zero, and A is three wide. Nothing is under B, so it rests on the floor, at y equals zero.

- Next door to A
- x = 0 + 3 = 3
- On the floor: y = 0

```
B at x = 3, y = 0
```

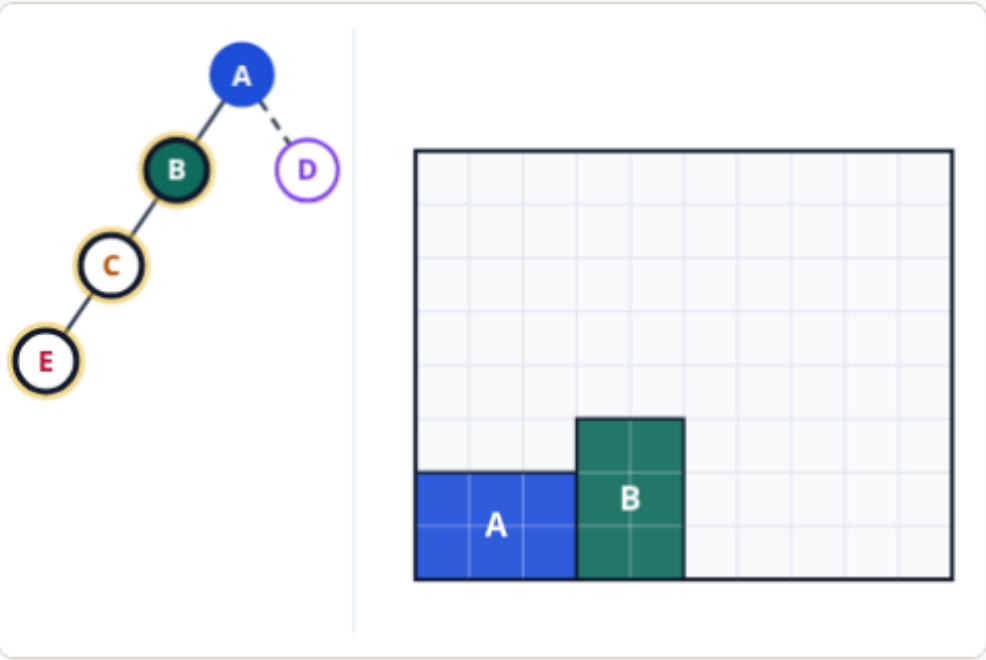
B goes next door, at x = 3

The walk: A, B, C, E, then D

B*-TREE

The walk: A, B, C, E, then D

Step 2 / 5 · left-chain · module02-03-bstar-tree



Explanation

We place the blocks one at a time, in a fixed walk order. Place a node, then everything down its left side, and only then its right child. So the walk goes A, B, C and E, and D comes last.

- Place a node
- then everything down its left side
- only then its right child

```
walk: A, B, C, E, D
```

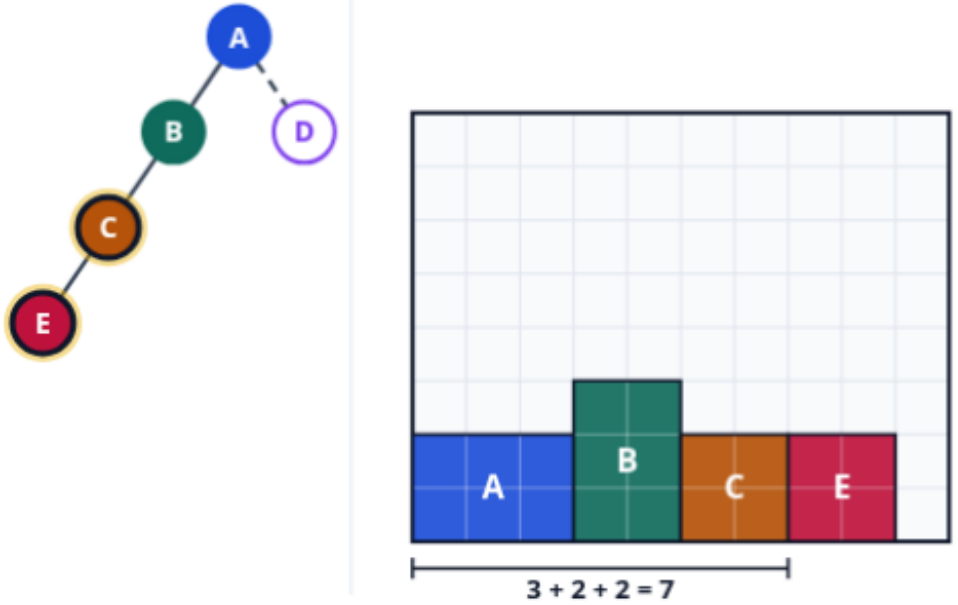
The walk: A, B, C, E, then D

C and E go next door

B*-TREE

C and E go next door

Step 3 / 5 · right-d · module02-03-bstar-tree



Explanation

C is B's left child, so it goes next door to B, at x equals five. E is C's left child, so it lands next door to C, at x equals seven. The widths simply add up: three, plus two, plus two, makes seven.

- C at x = 5
- E at x = 7
- 3 + 2 + 2 = 7

```
C at x = 5
E at x = 7
```

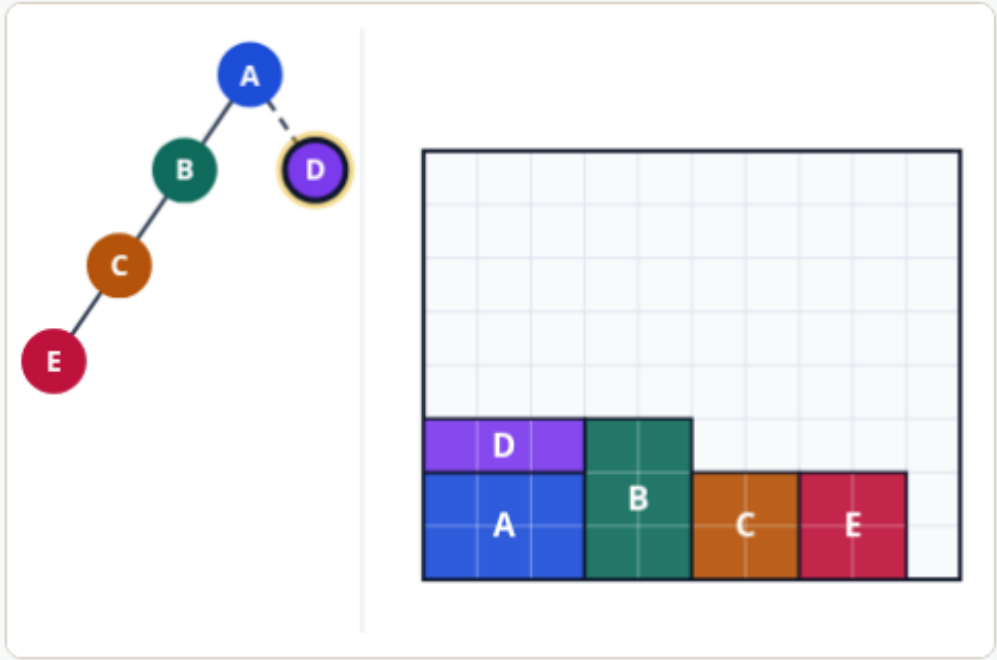
C and E go next door

D goes upstairs, on A

B*-TREE

D goes upstairs, on A

Step 4 / 5 · legal · module02-03-bstar-tree



Explanation

The left side is done, so the walk comes back to D, A's right child. D goes upstairs, so it starts at x equals zero, like A. It sits on top of A, at y equals two.

- D: A's right child
- Upstairs: x = 0
- On top of A: y = 2

```
D at x = 0, y = 2
```

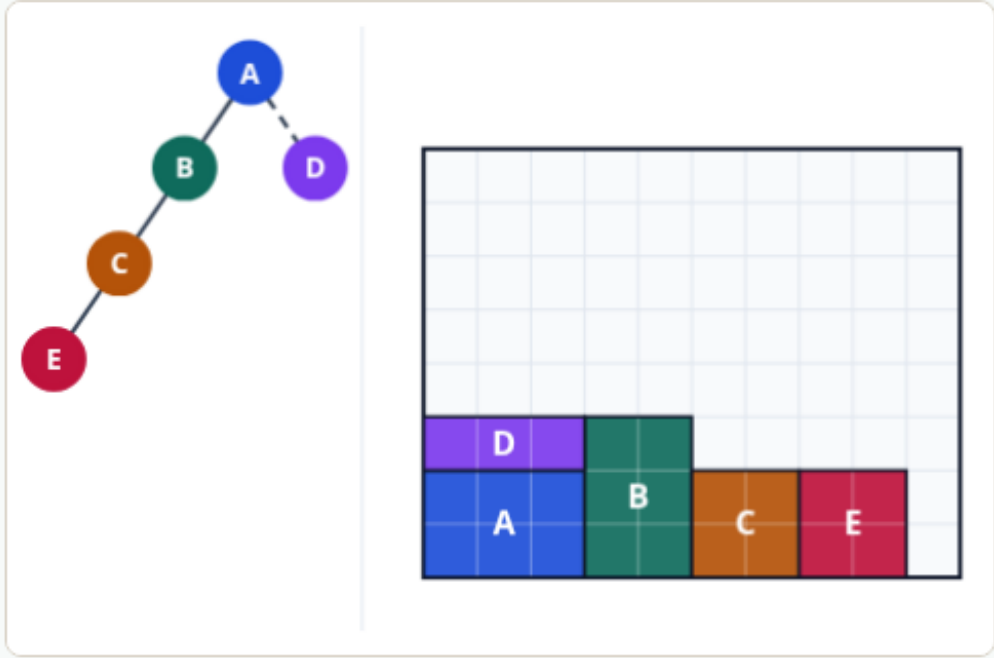
D goes upstairs, on A

Five blocks placed

B*-TREE

Five blocks placed

Step 5 / 5 · takeaway · module02-03-bstar-tree



Explanation

All five blocks are placed, and every one is inside the outline. Pause and guess: if we swap A and C in the tree, does D still land at y equals two?

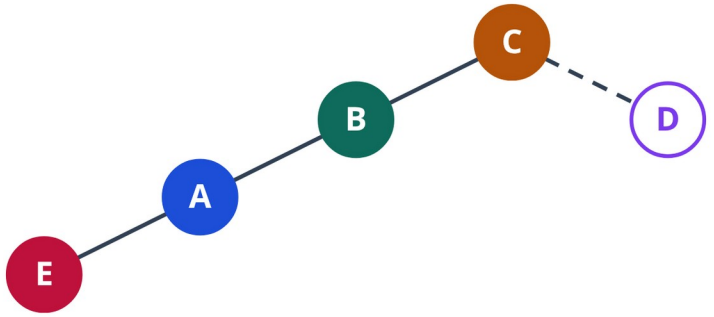
- All five inside the outline
- Swap A and C:
- is D still at y = 2?

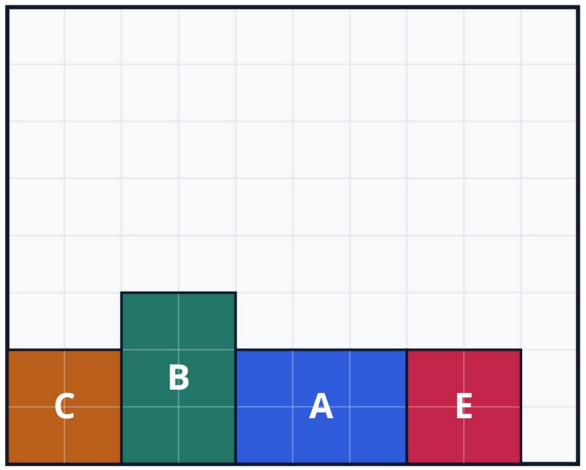
```
all 5 placed
```

Five blocks placed

Swap A and C

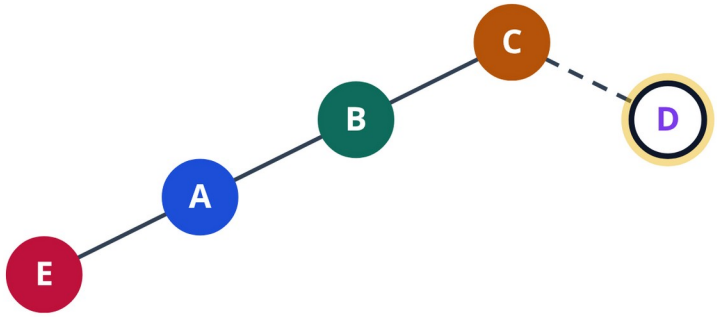
- No: D lands one square higher
- Here is the tree with A and C swapped: C is the root, and A takes C's old place
- C, B, A and E still go next door, one after the other

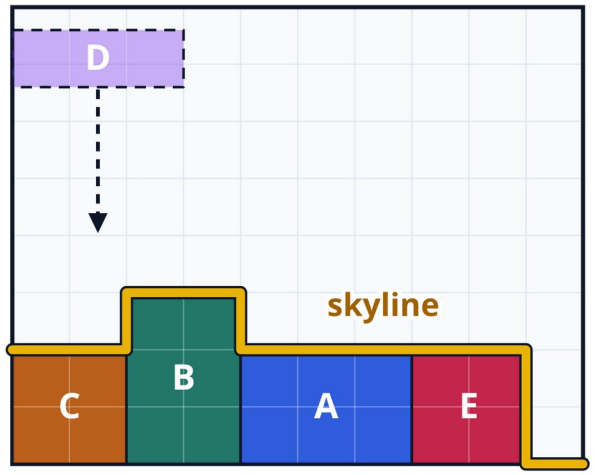




The skyline

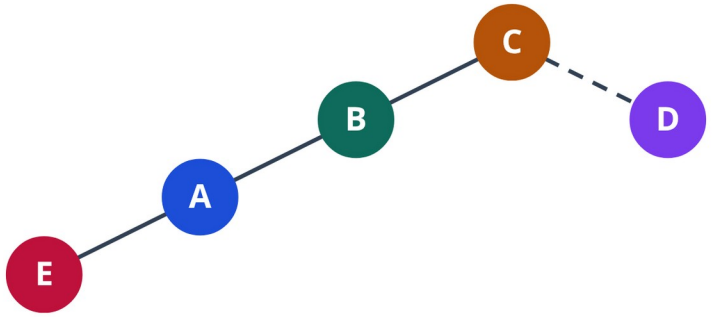
- Now picture the blocks as a city, seen from far away
- The top edge of everything placed so far is the skyline
- A new block drops from the sky, like a Tetris piece, until it touches the skyline

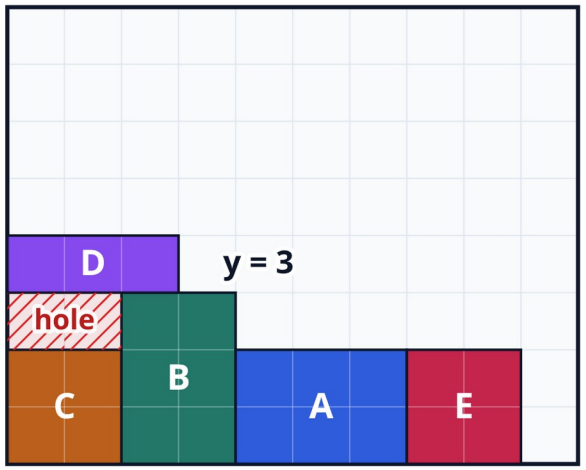




D lands on B

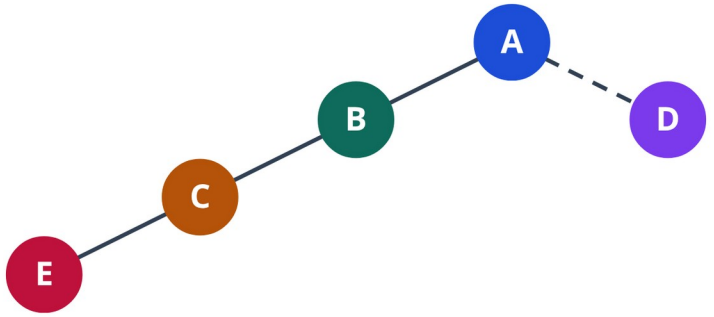
- D is three wide, so it hangs over C and over part of B
- B is three tall, so it stops D at y equals three
- That leaves a hole of two empty squares above C

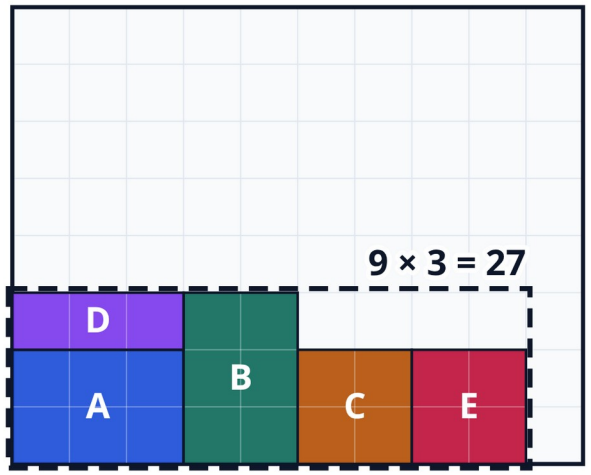




The bounding box

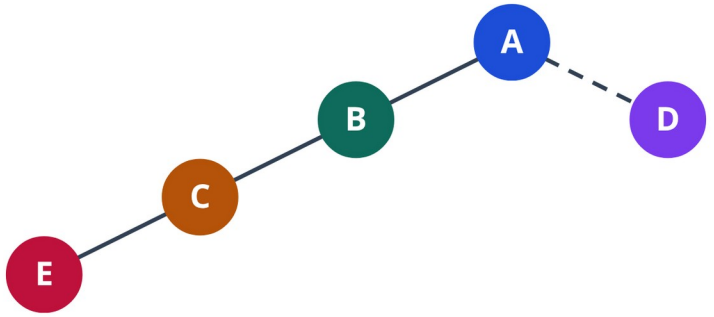
- So which floorplan is better?
- To compare, we draw the bounding box: the smallest rectangle around all the blocks
- For the first tree, it is nine wide and three tall: twenty-seven squares

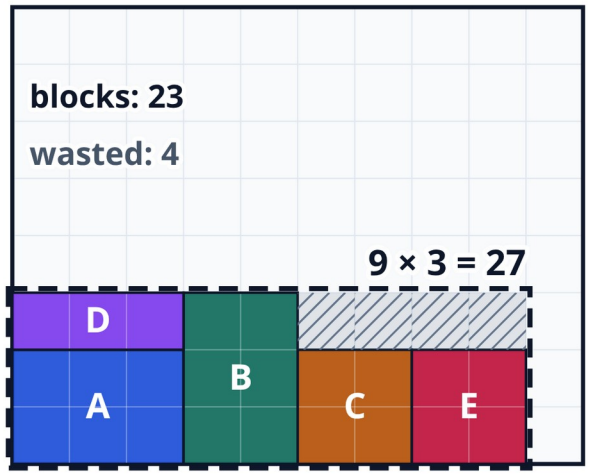




Wasted space

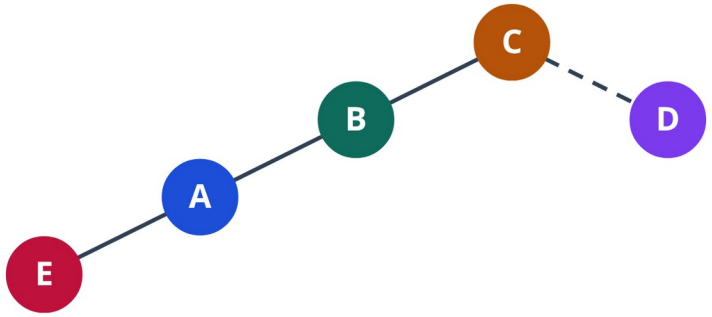
- The five blocks cover twenty-three squares
- The other four squares in the box are empty: that is wasted space
- Fewer wasted squares means a tighter floorplan

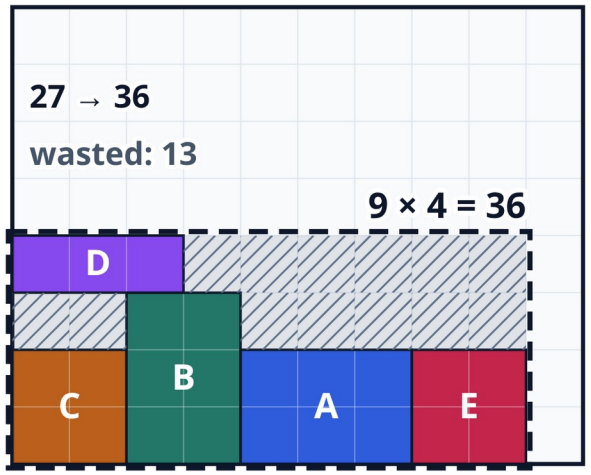




One swap, a worse score

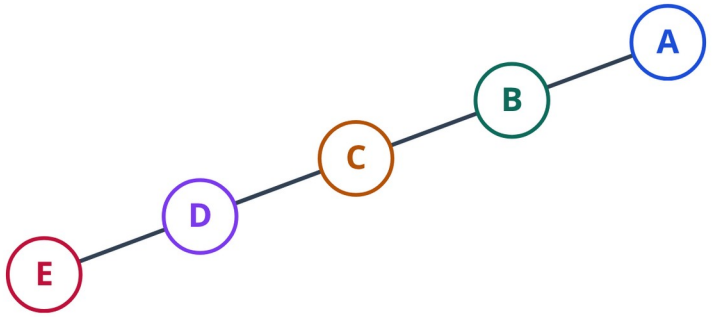
- The swapped tree needs a box nine by four: thirty-six squares, with thirteen wasted
- So one swap of two names changed the picture and the score
- Twenty-seven became thirty-six: the first tree is better

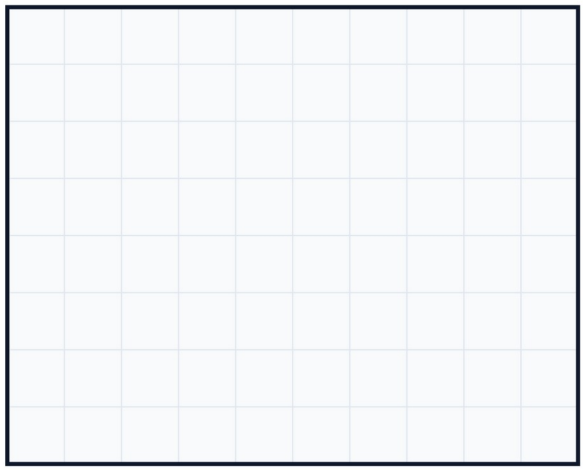




One long row

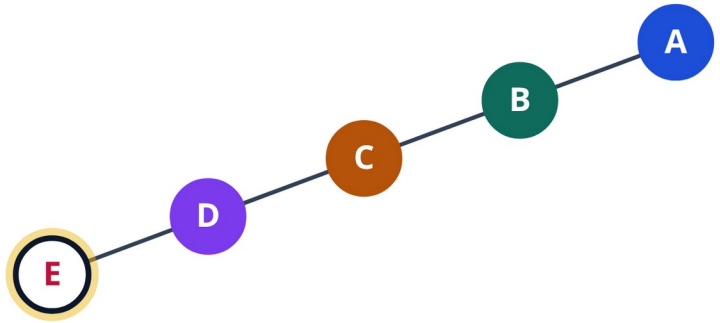
- Now take a tree where every child is a left child
- Pause and guess: will this long row fit inside the outline?

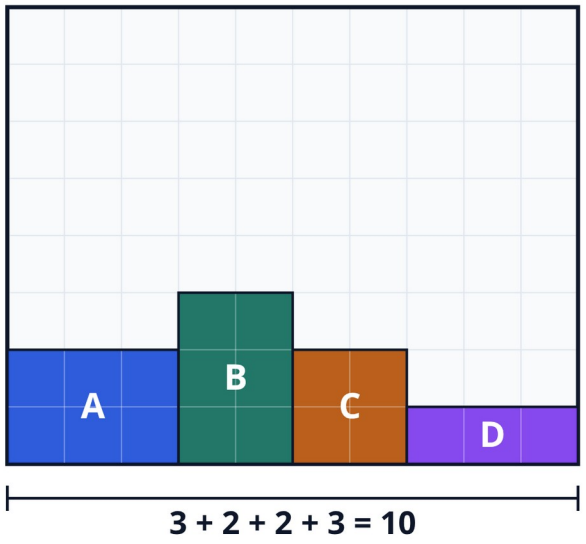




Every block next door

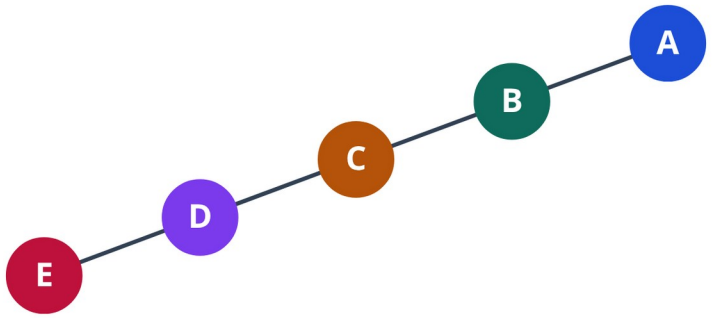
- No, it will not fit
- Every block goes next door: three, two, two and three make ten
- The outline is already full, and E is still waiting

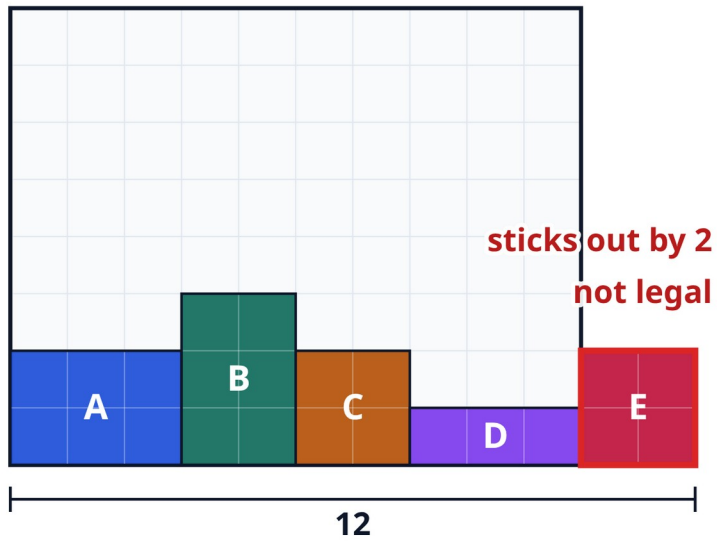




Sticking out

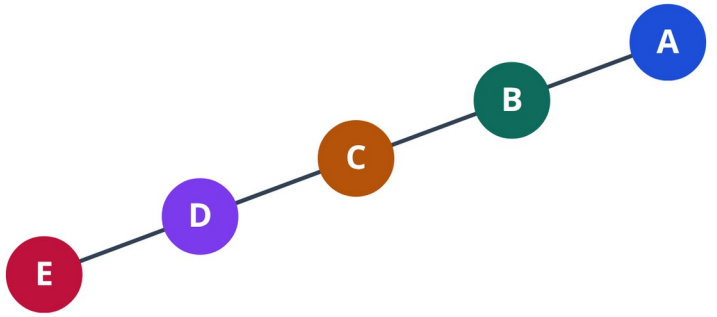
- E lands at x equals ten, so the row is twelve squares wide
- E sticks out of the outline by two squares
- A floorplan is legal only when every block is inside the outline
- So this row is not legal

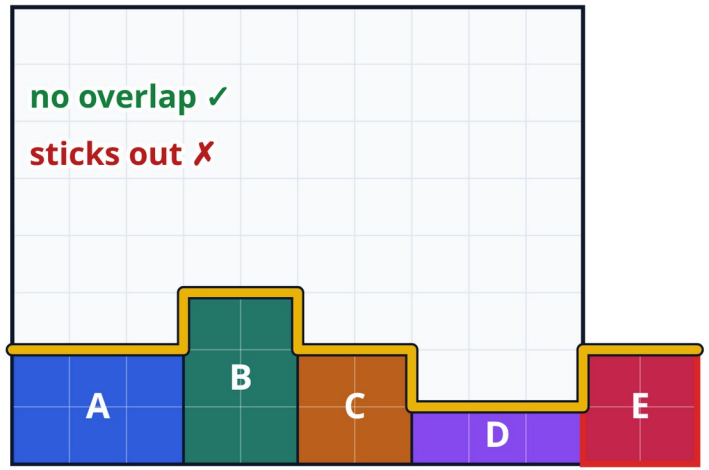




Never an overlap

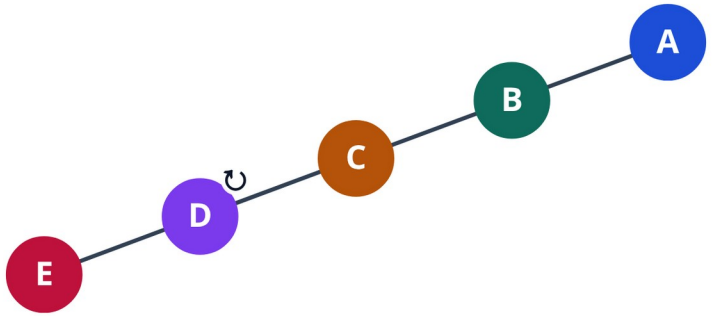
- Notice that the blocks never overlap
- Each block drops onto the skyline, so it can never land inside another one
- With a B-star tree, sticking out is the only way to fail

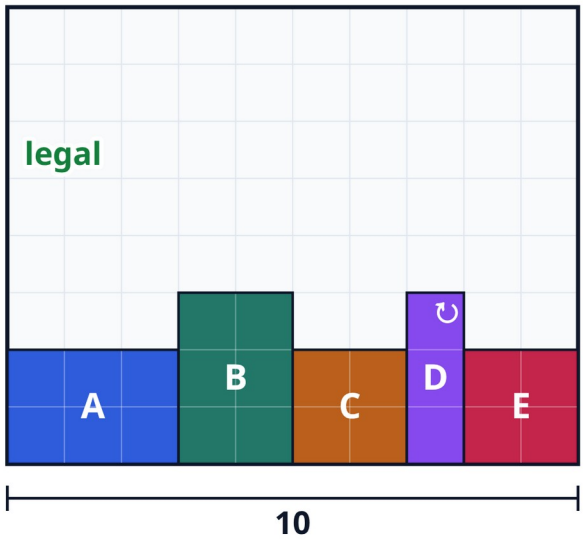




Turn one block

- We can turn a block on its side, so its width and height swap
- Turn D, and it becomes one wide and three tall
- Now the row is ten wide, so it fits exactly, and it is legal again



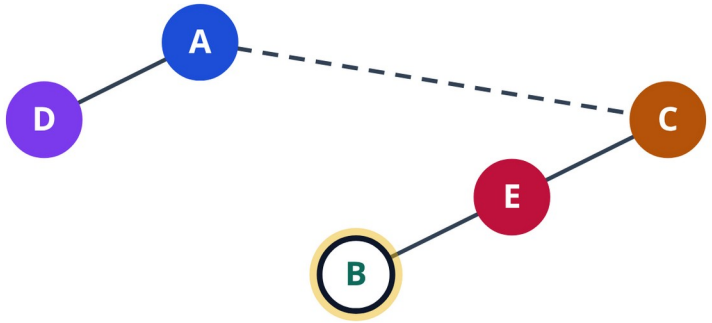


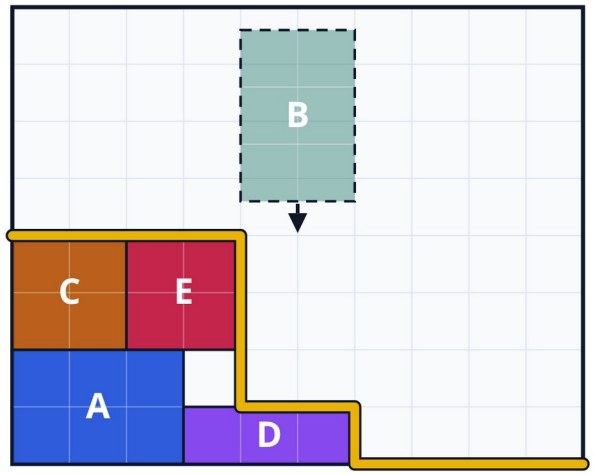
Three small changes

- Turning is one of three small changes
- You can also swap two names in the tree, as we did with A and C
- Or move a leaf, a block with no children, to a free spot in the tree
- Each change gives a new floorplan, with a new score
- Pause and guess: can any tree beat twenty-seven squares, and what is the smallest box?

The tightest floorplan

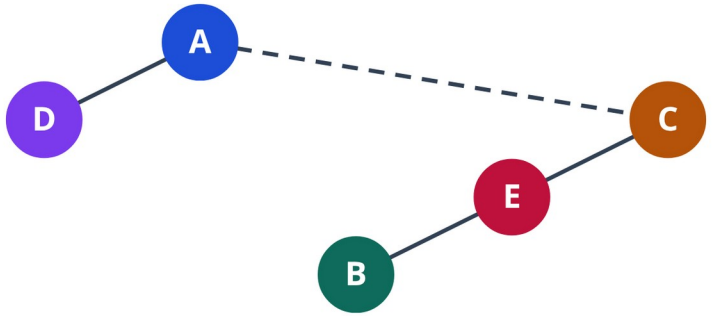
- Yes: many trees beat twenty-seven, and the best need only twenty-four squares
- Here is one of the best trees
- Its walk places A, D, C and E, and B comes last

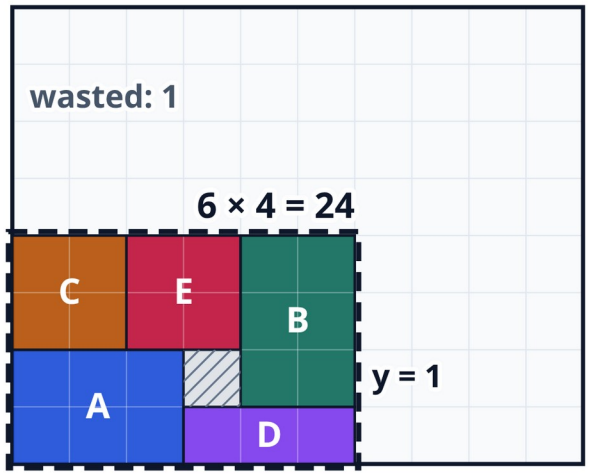




B is lifted by the skyline

- Under B, the skyline is the top of D, so B lands at y equals one
- It fills the gap, and the box is six wide and four tall: twenty-four squares
- Only one square is wasted





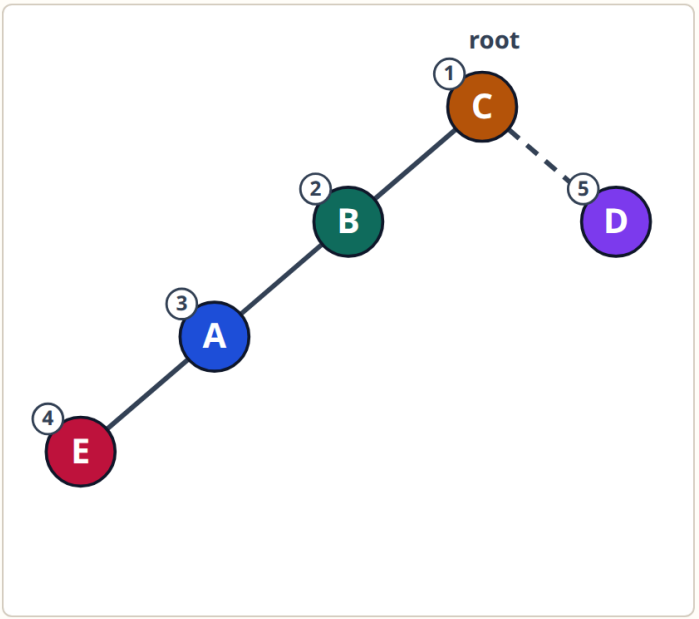
Why not twenty-three?

- Could a box of twenty-three squares, with nothing wasted, hold all five blocks?
- Twenty-three squares only make a rectangle one square thick
- Block B is too thick for that, so twenty-four is the best possible

Browser lab track

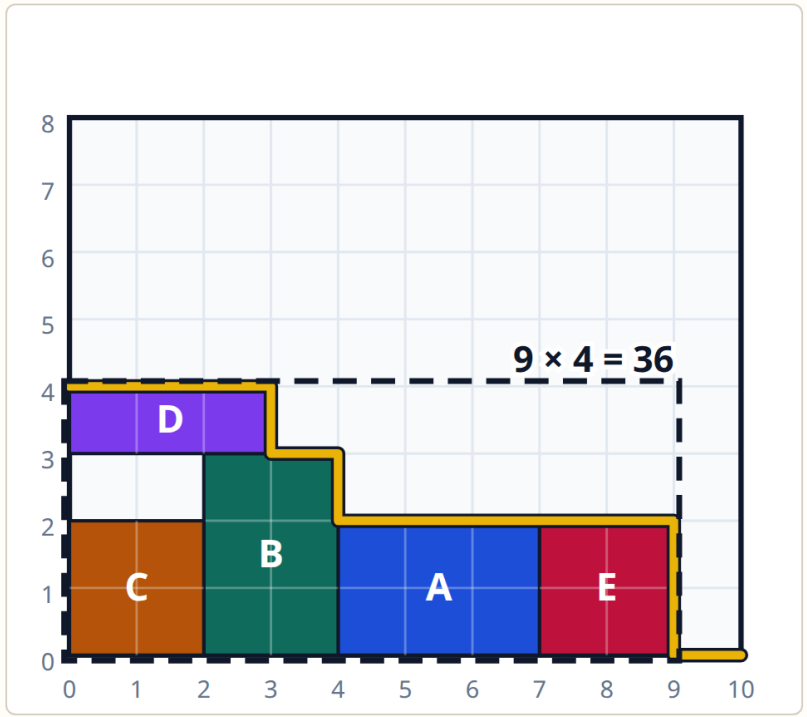
Tree and floorplan

The tree



solid line = next door (left child)
dashed line = upstairs (right child)
small number = walk order

The floorplan (outline 10 × 8)



yellow line = skyline
dashed box = bounding box

Browser lab starter

Implement track

wsl — module02-03-bstar-tree/examples/.

```
# Track A — real Unix
# real shell session (Track A)

$ python3 bstar_check.py --show | grep PASS | head -6
PASS  golden A(B(C(E,-),-),D): A (0,0) B (3,0) C (5,0) D (0,2) E (7,0)
PASS  golden: the module list in reverse order gives the same pack
PASS  golden: the pack keeps each block's w and h
PASS  golden: pack_bstar does not change the tree or the module list you pass
PASS  golden: bounding box 9x3 = 27
PASS  golden: legal in the 10x8 outline

$ python3 bstar_check.py --impl wrong/right_first.py | grep FAIL
FAIL  lift C(B(A(E,-),-),D): D lands at (0,3), not (0,2), because B is 3 high
FAIL  order A(B(C(-,E),D),-): left subtree before right child gives a 7x5 box
FAIL  best A(D,C(E(B,-),-)): B is lifted onto D, bounding box 6x4 = 24
FAIL  all 5040 labelled trees on A to E: your pack equals the reference pack
25 PASS, 4 FAIL, 0 SKIP
```

Real shell session

Implement track — try these

```
# print every correct answer
python3 examples/bstar_check.py --show

# check your own functions
python3 examples/bstar_check.py --impl my_bstar.py
```

Remember

- Left child: next door
- Right child: upstairs, with the same left edge
- Walk the tree: the node, its whole left side, then its right child
- Each block drops until it touches the skyline
- Score a floorplan with its bounding box, and count the wasted squares
- A tree never overlaps, but it can stick out, so check the outline

Your turn

- Pick a track: the browser lab, Python, or both
- Tick each box of the checklist, then take the short quiz
- Next comes the sequence pair: two lists of names instead of a tree
- Pause and guess: can two lists describe our best floorplan, with twenty-four squares?
- Yes: two lists are enough
- That's the sequence pair, in the next lab